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Group 30 — DSCI550 — April 9, 2026

Identifying NHL Player Archetypes Using K-Means Clustering and NHL EDGE Tracking Data

Introduction

In professional hockey, coaches must make rapid deployment decisions across distinct game situations such as even strength, power play, and penalty kill, each demanding a different balance of skill. While coaching wisdom provides heuristics, quantitative research and applied unsupervised learning to identify player archetypes from tracking data and evaluating situational effectiveness provides deeper insight to the game. This project addresses that gap using K-Means clustering on individual NHL statistics to classify skaters into player types and assess each type's on-ice impact. The nine initial input features capture offensive production, physicality, puck management, and athleticism (e.g., goalsPer60, hitsPer60, topShotSpeed). Output variables are on-ice goals-for and goals-against per 60 minutes at each game state. We hypothesize that offensive archetypes will yield stronger goal-for rates, especially on the power play while defensive types will demonstrate lower goals-against rates, especially on the penalty kill.

Project Idea and Motivation

The project idea was decided based on our interest in what different hockey play styles might look like when defined quantitatively. NHL lineup decisions are often intuition-based, with coaches often citing the "eye test" as reasoning behind their choices. By clustering players into types and measuring on-ice impact, we provide a quantitative basis for deployment. We use three seasons of data, as multiple years of data best predict future sports performance [2]. This work could benefit coaches, analysts, and fans seeking data-driven insight into player composition and team success, potentially giving fans specifically interested in player deployment a way to analyze coaches' decisions quantitatively.

Description of Data

We collected statistics from the last three NHL regular seasons (2023–24 through 2025–26) via the public NHL API [1] using the nhl-api-py Python module, scraping six report types with a paginated query system. We additionally fetched NHL Edge tracking data (skating speed, shot speed, zone time, distance skated) for each player-season using a multithreaded pipeline with 10

concurrent workers. Nested shot location data was parsed to engineer features for high/mid/long-danger shot counts and shots from specific ice areas (crease, low slot, high slot). The raw dataset contained ~2,754 player-season entries across 100+ columns.

Cleaning removed 22 null entries and filtered players with fewer than 10 games played. Counting stats were converted to rates: per-60-minute for overall and even-strength stats, per-2-minute for power play and penalty kill. The final dataset contains 2,299 rows and 171 columns.

Description of Models

We use K-Means [4] clustering via the python sci-kit-learn module to group skaters into player types, splitting forwards and defensemen into separate problems due to fundamentally different role distributions (e.g., defensemen average 4.58 blockedShotsPer60 vs. 2.20 for forwards; forwards average 0.77 goalsPer60 vs. 0.21 for defensemen).

Variable selection began with 100+ columns. Correlation analysis (Figures 1–2) guided our choices: goals and shots per 60 are positively correlated (r=0.66 forwards, r=0.58 defensemen) but sufficiently distinct for our initial tests; offensive and defensive metrics are negatively correlated (r=-0.36), confirming player specialization; skating and shot speed show low correlation with counting stats (r<0.20), contributing independent athletic information. We excluded redundant variables like pointsPer60 and totalShotAttemptsPer60 due to multicollinearity [3]. The final nine features are: goalsPer60, assistsPer60, blockedShotsPer60, hitsPer60, giveawaysPer60, takeawaysPer60, shotsPer60, skatingSpeed.speedMax, and topShotSpeed. Features were standardized via StandardScaler, and optimal k was chosen (k=3–8) by maximizing silhouette score.

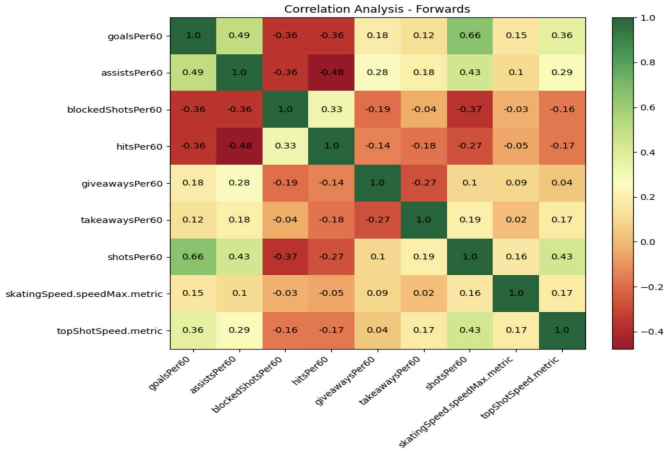


Figure 1: Correlation heatmap of clustering variables for forwards (n=1,504).

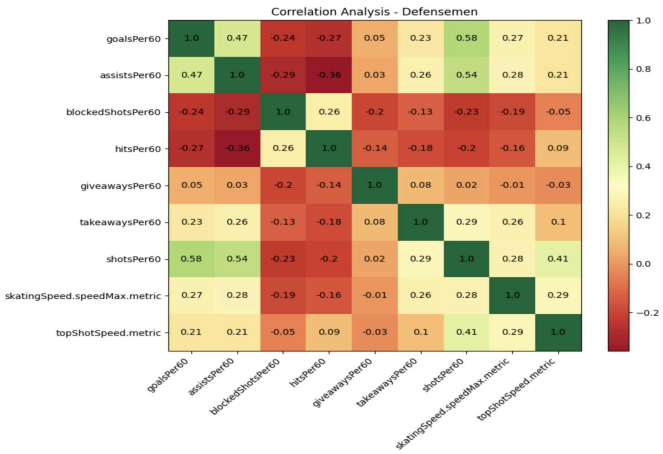


Figure 2: Correlation heatmap of clustering variables for defensemen (n=795).

Preliminary Results

K-Means produced 4 forward clusters and 3 defenseman clusters. Forward archetypes include: star offensive contributors (elevated goalsPer60, shotsPer60, assistsPer60) as cluster 2, balanced forwards as cluster 0, physical grinders (high hitsPer60, low offensive output) as cluster 3, and shutdown two-way forwards (low giveaways and high takeaways) as cluster 1. Defenseman clusters separate offensive (cluster 0), shutdown (high blocks/hits, cluster 2), and balanced types (cluster 1).

On-ice goal rates were computed per cluster at even strength, power play, and penalty kill (Figure 3). Initial results show meaningful situational differences by player type, supporting our hypothesis. In figures 5 and 6 we can clearly see that specific player types perform better in certain situations, i.e. shutdown defenders struggle on the power play, but thrive on the penalty kill. The final report will incorporate expected goals share and evaluate whether average speed metrics and shot location features improve clustering.

Forwards Cluster Profiles (Mean Rates):

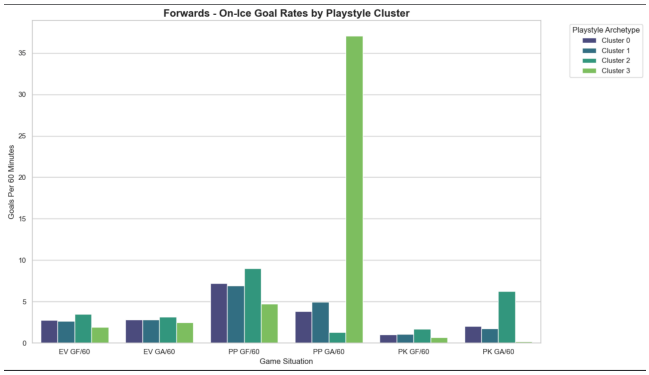
Cluster	goalsPer60	assistsPer60	blockedShotsPer60	hitsPer60	giveawaysPer60	takeawaysPer60	shotsPer60	skatingSpeed.speedMax.metric	topShotSpeed.metric
0	0.77	1.06	1.96	4.00	2.83	0.90	6.29	35.91	138.40
1	0.74	0.96	2.17	5.15	1.35	1.82	6.71	35.83	141.96
2	1.27	1.64	1.55	3.18	2.76	1.28	9.02	36.55	146.77
3	0.42	0.58	3.06	11.71	2.00	0.88	5.04	35.81	134.21

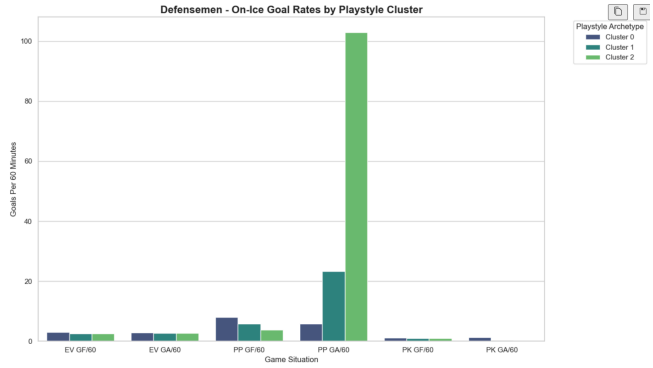
Figure 3: Cluster attribute averages for Forward clusters.

Defensemen Cluster Profiles (Mean Rates):

Cluster	goalsPer60	assistsPer60	blockedShotsPer60	hitsPer60	giveawaysPer60	takeawaysPer60	shotsPer60	skatingSpeed.speedMax.metric	topShotSpeed.metric
0	0.37	1.13	4.04	2.59	2.64	1.14	5.30	36.03	153.03
1	0.15	0.64	4.25	3.06	3.04	0.88	3.31	35.02	144.15
2	0.12	0.53	5.78	6.51	2.00	0.69	3.36	34.66	148.46

Figure 4: Cluster attribute averages for Defensemen clusters.





Figures 5, 6: GF/60 and GA/60 by cluster at even strength, power play, and penalty kill for forwards and defensemen, respectively.

References

[1] Zmalski, "NHL-API-Reference," GitHub, 2024. https://github.com/Zmalski/NHL-API-Reference.

[2] A. Chamberlain, "Hot Hand or Hard Data: Is Recent Performance Weighted Enough in Pitching Projections?," FanGraphs, 2023. https://fantasy.fangraphs.com/hot-hand-or-hard-data-is-recent-performance-weighted-enough-in-pitching-projections/.

[3] Singh, V. (2024, October 28). "Multicollinearity in Regression: A Guide for Data Scientists." Datacamp.com; DataCamp. https://www.datacamp.com/tutorial/multicollinearity.

[4] "Leveraging K-Means Clustering for Sports Analytics." *FlyRank*, 25 Apr. 2025, flyrank.zendesk.com/hc/en-us/articles/26277513527698-Leveraging-K-Means-Clustering-for-Sports-Analytics.